

ABHINAYA S B

Security, Privacy & Online Safety Researcher Bridging Human-Centered Methods and AI-Driven Systems

Ph.D. Candidate, Dept. of Computer Science, North Carolina State University

Raleigh, NC | asrivid@ncsu.edu | abhinayasb.github.io | linkedin.com/in/sbabhinaya

Professional Summary

- 7+ years of experience in Security, Privacy & Online Safety Research, ML, NLP, and Software Engineering.
- Research leveraging **NLP**, multi-modal **LLMs**, and **human-centered** methods to study **online safety**, **privacy** and **scams** in emerging technologies including GenAI, VR, and voice assistants.
- Publications at top-tier **security**, **privacy** & **applied AI** venues (USENIX Security, IEEE S&P, PETS, ICWSM).

Education

North Carolina State University (NCSU), Raleigh, USA

Aug 2022 - Present (est. May 2027)

PhD in Computer Science; *Advised by Dr. Anupam Das (WSPR lab)*

CGPA: 4.00/4.00

National Institute of Technology, Tiruchirappalli, India

Jul 2015 - May 2019

Bachelor of Technology (Honours) in Computer Science and Engineering

CGPA: 9.74/10.00

Employment & Experience

Research Experience

Graduate Research Assistant, Security and Privacy Research Lab, University of Washington

May 2026 - Present

Supervisor: Prof. Franziska Roesner

- **Parent-aligned guardrails for children's generative AI use** (under preparation)
Evaluating safety compliance of children's GenAI tools, how well AI guardrail models enforce parents' custom safety preferences, and the strategies children use to bypass these custom restrictions.
- **Perspectives of Platforms, Parents, and Children on Safe GenAI for Children** (in submission)
Designed a mixed-methods study investigating parents' and children's needs around GenAI parental controls. Conducted a survey of parents, cooperative inquiry with children aged 6–11, and analyzed children's GenAI tools.

Graduate Research Assistant, WSPR lab, North Carolina State University

Aug 2022 - Present

Advisor: Prof. Anupam Das

- **Multimodal AI system for safety risk assessment in VR environments** (in submission)
Curated a multimodal Social VR dataset. Developed and evaluated multimodal LLM pipelines (speech, audio, and visual modalities) for VR safety risk detection. Prototyped an AI-assisted moderation tool with multimodal RAG and assessed human-AI collaboration UX through a user study with expert moderators.
- **Studying persuasive tactics of telephone-based scammers** (in submission)
Built an LLM-based automated system to label and characterize scammer-victim interactions, uncovered scammer narratives and persuasion tactics. Leveraged diarized call transcripts extracted from YouTube scambaiting videos. Developed a novel theoretical model to describe the complex dynamics of scammer-victim interactions.
- **Identifying scam video content using a multimodal, policy-grounded approach** (ICWSM, 2026, P2)
Systematically analyzed multimodal LLM-based approaches for scam video identification. Fine-tuned LLaVA-Video to provide interpretable, policy-grounded reasoning.
- **Designing usable privacy notices for voice interfaces** (PETS, 2026, P1)
Designed a mixed-methods study to identify the key design and contextual factors for effective privacy notices for voice interfaces. Used statistical analysis to evaluate prototype designs with high user preference and usability.

- **Analyzing ad prevalence, characteristics, and compliance in Alexa skills** (IEEE S&P, 2025, P4)
Developed an automated ad detection pipeline using a LoRA fine-tuned LLM using chain-of-thought prompting, identifying and characterizing various types of policy-violating advertisements on Amazon Alexa platform.
- **Understanding privacy expectations, concerns and behaviors of VR users** (PETS, 2025, P3)
Led an interview-based study with VR users to elicit their diverse perceptions regarding the data practices of VR companies, their privacy concerns with using VR, and the steps they took to protect their privacy.
- **Understanding parents’ perceptions and practices toward children’s S&P in VR** (IEEE S&P, 2024, P7)
Designed and conducted an interview-based study with parents of child VR users to extract insights about parents’ security and privacy (S&P) knowledge, perceptions, and practices towards addressing their children’s S&P.
- **Investigating harassment and safety mechanisms in VR** (USENIX Security, 2024, P5)
Designed and led a multi-perspective interview study with (i) VR users and (ii) VR developers to identify the usability and development challenges with existing VR safety controls.

Industry Experience

- **Engineer, Qualcomm India Pvt. Ltd.** Dec 2020 - Jun 2022
Designed a prototype to optimize memory for concurrent workloads on the AI-processor
Added testing support for 2 simulators, 1 emulator, 2 Linux and 1 Windows based platforms.
- **Associate Engineer, Qualcomm India Pvt. Ltd.** Jul 2019 - Nov 2020
Developed a suite of tests and tools for Qualcomm’s AI-processor that identified 5 critical HW issues; presented work at internal conferences (Qbuzz, ML Summit) with 3K+ attendees.
- **Interim Engineering Intern, Qualcomm India Pvt. Ltd.** May 2018 - Jul 2018
Analyzed Google’s Pixel Visual Core based image capture to derive architectural insights.

Skills

Programming	Python, C++, C, Java, Tensorflow, HTML/CSS, Javascript, Shell
Tools	LaTeX, Git, ollama, vllm, Gradio, MongoDB, Dedoose, Android Studio, MATLAB, Trace32, Zotero
Methods	Natural Language Processing, Clustering, Prompt Engineering, LLM Fine-tuning, Multimodal RAG, LLM Agents, Survey Design, Interview Design, Qualitative Analysis, Statistical Analysis
Courses	Privacy in the Digital Age, Computer & Network Security, Software Security, Human-Centered Research Methods, Cryptography, Cellular & Telephone Network Security, Human-Centered Security, Advanced Network Security, Algorithms, Computer Networks

Peer-Reviewed Publications

P1	Privacy by Voice: Designing Usable Privacy Notices for the Voice Interface Aafaq Sabir, <u>Abhinaya S.B.</u> , Dilawer Ahmed, Anupam Das Privacy Enhancing Technologies Symposium (PETS), Jul 2026.	[paper]
P2	Beyond Metadata: Multimodal, Policy-Aware Detection of YouTube Scam Videos Ummay Kulsum, Aafaq Sabir, <u>Abhinaya S.B.</u> , Anupam Das International AAAI Conference on Web and Social Media (ICWSM), May 2026.	[paper]
P3	“What are they gonna do with my data?”: Privacy Expectations, Concerns, and Behaviors in Virtual Reality <u>Abhinaya S.B.</u> , Abhishri Agrawal, Yaxing Yao, Yixin Zou, Anupam Das Privacy Enhancing Technologies Symposium (PETS), Jul 2025.	[paper]
P4	Analyzing Ad Prevalence, Characteristics, and Compliance in Alexa Skills Aafaq Sabir, <u>Abhinaya S.B.</u> , Dilawer Ahmed, Anupam Das IEEE Symposium on Security and Privacy, May 2025. (Acceptance Rate: 14.8%)	[paper]
P5	Enabling Developers, Protecting Users: Investigating Harassment & Safety in VR <u>Abhinaya S.B.</u> , Aafaq Sabir, Anupam Das USENIX Security Symposium, Aug 2024. (Acceptance Rate: 18.3%)	[paper]

P6

Unveiling Users’ Security and Privacy Concerns Regarding Smart Home IoT Products from Online Reviews

Taufiq Islam Protick, Aafaq Sabir, Abhinaya S.B., Aiden Bartlett, Anupam Das

ACM Journal on Computing and Sustainable Societies, Aug 2024.

[paper]

P7

Understanding Parents’ Perceptions and Practices Toward Children’s Security and Privacy in Virtual Reality

Jiaxun Cao, Abhinaya S.B., Anupam Das, Pardis Emami-Naeini

IEEE Symposium on Security and Privacy, May 2024. (Acceptance Rate: 17.8%)

[paper]

P8

A Question Answering and Quiz Generation Chatbot for Education

A S Sreelakshmi, Abhinaya S.B., Aishwarya Nair, S Jaya Nirmala

Grace Hopper Celebration India (GHCI), Nov 2019.

[paper]

P9

A Survey of Techniques for improving Security of GPUs

Sparsh Mittal, Abhinaya S.B., Manish Reddy, Irfan Ali

Journal of Hardware and Systems Security, Sep 2018.

[paper]

Posters

P1

Virtual Adventures, Real Challenges: Analyzing Harassment Controls in VR

Abhinaya S.B., Aafaq Sabir, Anupam Das

USENIX Security Symposium, Aug 2023.

[poster]

P2

Protego Totalum! Mitigating Harassment in Virtual Reality

Abhinaya S.B., Aafaq Sabir, Anupam Das

Graduate Student Research Symposium, NC State University, Apr 2023.

[abstract]

Service

- Program Committee Member: USENIX Security (2027)
- Artifact Evaluation Committee: USENIX Security (2023, 2024, 2025)
- External Reviewer: ICWSM (2027), IEEE S&P (2024, 2025), USENIX Security (2025, 2026), NDSS (2026), CCS (2023), ACSAC (2023), CSCW (2024), USEC (2024)

Awards & Honors

- **SOUPS’25 Student Grant:** Awarded 400 USD to attend Symposium on Usable Privacy and Security in Aug 2025
- **PETS’25 Travel Stipend:** Awarded 650 USD to attend Privacy Enhancing Technologies Symposium in Jul 2025
- **USENIX Security’24 Diversity Grant:** Awarded 625 USD to attend USENIX Security Symposium in Aug 2024
- **Graduate Merit Award:** Awarded 3000 USD by the College of Engineering, NCSU in Mar 2024
- **RSA Security Scholar:** Selected as a Security Scholar to attend the RSA conference, in Nov 2023
- **USENIX Security’23 Diversity Grant:** Awarded 850 USD to attend USENIX Security Symposium in Aug 2023
- **GREPSEC Scholar:** Awarded 970 USD to attend the GREPSEC VI workshop in Aug 2023
- **Graduate Enhancement Award:** Awarded 2850 USD by the College of Engineering, NCSU in Apr 2023
- **North Carolina State University Graduate Fellowship:** Awarded 4000 USD by NCSU in Feb 2022

Invited Talks & Lectures

- **“What are they gonna do with my data?”: Privacy Expectations, Concerns, and Behaviors in VR**
Conference presentation, 25th Privacy Enhancing Technologies Symposium (PETS), Washington, DC, USA Jul 2025
- **Towards Better Privacy & Safety for VR Users**
Invited Talk, Triangle Area Privacy and Security Day (TAPS), Duke University Oct 2024

- **Human-Centered Security & Privacy Research for Virtual Reality**
Classroom lecture, CSC 801: Seminar in Computer Science, North Carolina State University Sep 2024
- **Panel discussion for early-stage PhD students**
Panelist, CSC 801: Seminar in Computer Science, North Carolina State University Sep 2024
- **Enabling Users, Protecting Developers: Investigating Harassment and Safety in VR**
Conference presentation, 33rd USENIX Security Symposium, Philadelphia, PA, USA Aug 2024
- **Enabling Users, Protecting Developers: Investigating Harassment and Safety in VR**
Invited talk, 3rd annual North Carolina Cybersecurity Symposium Feb 2024

Select Media Coverage

- NC State Alumni Magazine – *“Headset Harms”*
- NC State News – *“VR Poses Privacy Risks for Kids. A New Study Finds Parents Aren’t as Worried as They Should Be.”*
- Duke Today – *“Parents Underestimate the Privacy Risks Kids Face in Virtual Reality.”*

Leadership

- Interview Prep Nights Committee, Women in Computer Science (WiCS), NCSU Jul 2023 - May 2024
- Vice President, Public Relations, Qonverse Toastmasters Club, Bangalore Jul 2021 - Dec 2021
- Secretary, Qonverse Toastmasters Club, Bangalore Jan 2022 - Jun 2022

Mentoring

- Abhishri Agrawal, UNC-Chapel Hill
- Dinesh Karnati, North Carolina State University
- Tarun Puvvala, North Carolina School of Science and Mathematics
- Noah Chang, North Carolina School of Science and Mathematics